MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG4DSEECT204 ROBOTICS

Duration: 2 hours **Maximum Marks: 70**

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability.*

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	Identify which of the following best defines a robot? a) A mechanical toy b) A programmable machine capable of carrying out tasks autonomously c) A computer system d) An electronic calculator	U	1
2	Select the first industrial robot was introduced mainly for: a) Medical applications b) Space exploration c) Repetitive manufacturing tasks d) Domestic use	U	1
3	Identify the component which provides movement to a robot? a) Controller b) Sensor c) Actuator d) End effector	K	1
4	Robots classified based on structure include: a) Mobile and aerial robots b) Cartesian, cylindrical, spherical robots c) Industrial and service robots d) Autonomous and semi-autonomous robots	U	1
5	Identify the use of Industrial robots a) Household cleaning b) Entertainment c) Assembly and welding operations d) Educational purposes	A	2
6	Map what degrees of Freedom (DOF) in a robot refers to: a) Speed of motion b) Number of independent movements c) Size of the robot d) Weight of the robot	A	2
7	Write the suitable motor commonly used for precise angular movement? a) DC motor b) Linear motor c) Stepper motor d) Hydraulic motor	A	2
8	Identify the Use of Pneumatic systems a) Electrical energy b) Hydraulic oil c) Compressed air d) Magnetic energy	U	2

9	Identify the Use of Sensors in robots a) Store energy b) Provide perception of environment c) Increase speed d) Reduce size	U	3
10	Identify the Use of Ultrasonic sensors: a) Temperature measurement b) Distance measurement c) Color detection d) Pressure sensing	U	3
11	Define how feedback control system improves: a) Weight b) Accuracy and stability c) Cost d) Appearance	K	3
12	Identify the platform is commonly used for robotics programming? a) MS Word b) Arduino c) Excel d) Photoshop	A	3
13	Write one application of Spot welding robots : a) Hospitals b) Space stations c) Automobile industries d) Agriculture	A	4
14	Identify what actually medical robots are primarily used for: a) Manufacturing b) Surgery and rehabilitation c) Entertainment d) Mining	A	4
15	Contrast the Human-Robot Interaction mainly focuses on: a) Robot speed b) Safe and efficient collaboration c) Robot size d) Power consumption	An	4
16	Write the ethical issues in robotics mainly relate to: a) Color of robots b) Battery type c) Job displacement and safety d) Robot programming language	A	4

1 * 16 = 16

Part B

Short Answer Questions

Answer any **6** Questions

Each question carries 5 marks

17	Describe robotics and explain its importance.	U	1
18	Explain the basic components of a robotic system.	U	1
19	Write the concept of Degrees of Freedom in robots.	A	2
20	Write about DC motor and stepper motor used in robotics.	A	2
21	Outline the role of sensors in robotic perception.	An	3
22	Illustrate feedback control system in robots.	An	3
23	Write short notes on service robots.	A	4
24	Contrast different ethical challenges in robotics.	An	4

6 * 5 = 30

Part C

Essay Questions

Answer any **2** Questions

Each question carries 12 marks

25	Explain the evolution, definition, and classification of robots in detail.	U	1
26	Write about robot kinematics, workspace, and actuation systems with suitable examples.	A	2
27	Illustrate sensors, control systems, and programming concepts used in modern robots.	An	3
28	Compare industrial, medical, and advanced robotic applications along with future trends and ethical issues.	An	4

2 * 12 = 24